Multi-Modal Machine Learning Approach for Threat Detection in WhatsApp and Telegram: Spam Links, Document Based Malware, Chat Behavior, and Sensitive Content Analysis

Detection of Malware- Infected Documents in Messaging Apps Using Sandbox-Based Behavior Analysis

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Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

This project proposal focuses on improving the security of messaging platforms by enabling users to check for malware-infected documents through sandbox-based behavior analysis. The primary objective is to design a secure and lightweight detection mechanism where users can manually upload or forward documents such as PDFs, DOCs, and APKs received via WhatsApp into the application for verification. Once uploaded, the document is executed in a sandbox environment that simulates real-world behavior in a controlled setting. The system then analyzes behavioral indicators such as file modifications, registry access, API calls, and network activity to determine if the file is malicious. If the document is found to be safe, the application will generate an alert notifying the user that it can be opened without risk. Conversely, if malicious activity is detected, the system will warn the user not to open the file. For cases where the file must still be accessed despite potential risks, the document can be safely opened and interacted with inside the sandbox without compromising the user's device. This research addresses limitations in traditional antivirus methods, which often rely on static scans and fail to detect hidden threats such as macro viruses, embedded scripts, and polymorphic malware. The methodology includes sandbox-based dynamic analysis, machine learning classification of malicious behaviors, and integration with a simple mobile interface for user interaction. By combining dynamic behavior monitoring with user-controlled file submission, the proposed solution aims to provide a proactive, accurate, and user-friendly framework for securing document sharing in messaging platforms.

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List of Abbreviations

ML	-	Machine Learning
API	-	Application Programming Interface
PDF	-	Portable Document Format
DOC	-	Microsoft Word Document
APK	-	Android Package Kit
UI	-	User Interface
VM	-	Virtual Machine
OP	-	Operating System
TP	-	True Positive
FP	-	False Positive
ROC	-	Receiver Operating Characteristic
AUC	-	Area Under Curve

I. INTRODUCTION

III.1 Background and Literature Review

The rapid growth of social media and messaging platforms such as WhatsApp and Telegram has reshaped digital communication by allowing instant information sharing among a wide range of users. Alongside these benefits, there is now greater exposure to cyber threats, particularly from malicious files delivered through shared documents. Attackers often exploit common formats such as PDFs, DOCs, and APKs to spread malware that can compromise devices, extract sensitive data, or launch further attacks.

Traditional antivirus systems rely mainly on static signature-based detection. While these methods can identify known threats, they are less effective against polymorphic malware, macro viruses, and files containing obfuscated code. To address these limitations, researchers have increasingly turned to dynamic and behavior-based analysis, which is better suited for uncovering hidden or evolving threats within messaging platforms.

Sandbox-based detection is a promising approach in this area. By executing suspicious files in a controlled environment, sandboxes capture runtime behaviors such as API calls, file modifications, registry changes, and network activity. These indicators provide richer evidence than static analysis and make it possible to detect zero-day and polymorphic malware. Although sandboxing has long been used in enterprise security, its integration into real-time mobile messaging platforms remains uncommon. A major challenge lies in creating sandbox systems that are lightweight and efficient enough to operate on mobile devices without affecting usability.

To explore this gap, a systematic literature review (SLR) was conducted, covering 315 papers from IEEE Xplore, ScienceDirect, Wiley, Google Scholar, and other trusted sources. Using the PRISMA framework, ten key review papers were developed, grouped into themes such as sandbox-based dynamic analysis, multi-file-type malware detection, machine learning–based classification, and mobile security. Findings confirm that while malware detection has advanced significantly, very few solutions apply sandbox-based behavior analysis directly within messaging environments.

This study addresses that limitation by proposing a sandbox-integrated mobile application that

inspects documents shared through WhatsApp and Telegram. The system allows a received file to be redirected into a sandbox before the user can open it. Inside this environment, runtime behavior is observed and features such as API calls, registry activity, and file system changes are extracted. A machine learning model then classifies the file as benign or malicious.

If the file is safe, the user is informed that it can be opened without risk. If the file is malicious, a warning is displayed, offering the user two options: either block the file or, if the content is essential, open it in a secure sandbox viewer where it cannot harm the device. This workflow ensures that children, who may be more likely to open harmful files, are protected from infection while still being able to access legitimate content when needed.

In summary, this research contributes a novel use of sandbox-based behavior analysis in messaging platforms to detect document-based malware. Unlike static antivirus scans, this method simulates real file execution, exposing threats such as macro viruses, embedded scripts, and polymorphic malware that often evade traditional detection. By embedding this capability within a child-safety–oriented mobile application, the work provides a practical, accurate, and user-friendly method for protecting WhatsApp and Telegram users from document-borne cyber threats in real time.

III.2 Research Gap

Although research in malware detection has advanced in recent years, major limitations remain when applying these methods to messaging platforms such as WhatsApp and Telegram.

Research [1] applies static analysis to detect malicious documents. While it demonstrates some success, static methods cannot reliably identify polymorphic malware or files containing obfuscated code. Research [2] uses machine learning to classify malware across several file types, but it does not combine this with sandbox-based behavior analysis. Without behavioral evidence, such systems often fail to uncover hidden threats like macro viruses or embedded scripts. Research [3] explores sandbox-based malware detection in enterprise contexts, yet these solutions are computationally heavy and not designed for lightweight, mobile-friendly deployment within real-time messaging ecosystems.

Taken together, these studies show a clear gap: there is no efficient, mobile-oriented system that integrates sandbox-based behavioral analysis directly into messaging workflows for detecting malicious documents. Existing methods are either restricted to static detection, omit sandboxing altogether, or rely on designs that are too resource-intensive for mobile devices.

The proposed solution addresses this need by developing a sandbox-based malware detection system tailored for WhatsApp and Telegram. Documents such as PDFs, DOCs, and APKs are executed in a secure sandbox environment, where runtime behaviors—including API calls, registry activity, file modifications, and network communication—are captured. These behavioral features are then classified using machine learning to determine whether a file is benign or malicious. If malicious, the system issues a warning and allows the file to be opened only in a secure sandbox viewer. This approach ensures accurate detection of sophisticated threats while remaining lightweight and practical for mobile messaging use.

Table I-1 - Research Gap in tabular view

Research Paper	Static Analysis	ML-Based Malware Classification	Sandbox-Based Detection	Messaging Platform Integration	Lightweight / Mobile-Friendly
Research [1]	Implemented	Not Implemented	Implemented	Not Implemented	Not Implemented
Research [2]	Not Implemented	Implemented	Not Implemented	Not Implemented	Not Implemented
Research [3]	Not Implemented	Implemented	Implemented	Not Implemented	Not Implemented
Proposed Solution	Implemented	Implemented	Implemented	Implemented	Implemented

The table highlights how earlier research has addressed parts of the malware detection problem but has left key areas unresolved. Research [1] combines static analysis with sandboxing but does not employ machine learning and is not integrated into messaging workflows. Research [2] demonstrates ML-based classification but excludes sandboxing and ignores the messaging platform context. Research [3] introduces sandbox-based detection but focuses on enterprise systems, which are resource-intensive and unsuitable for lightweight mobile environments.

By comparison, the proposed solution integrates **all five elements**: static analysis, ML-based malware classification, sandbox-based detection, lightweight design, and direct integration with messaging platforms. This combination makes it well suited for protecting children and other vulnerable users from document-based malware within WhatsApp and Telegram, while remaining efficient and practical for real-world mobile use.

III.3 Research Problem

The rise of messaging platforms such as WhatsApp and Telegram has created new opportunities for communication, particularly during the COVID-19 pandemic when children relied heavily on these platforms for learning and interaction. However, this increased usage has also exposed them to cybersecurity threats, one of the most critical being document-based malware. Attackers frequently embed malicious code in familiar file formats such as PDFs, DOCs, and APKs. Once opened, these files can execute hidden instructions, resulting in file system changes, registry modifications, or unauthorized network activity.

Conventional antivirus software relies mainly on static detection methods, which are limited in scope. These methods struggle to identify advanced threats such as polymorphic malware, obfuscated scripts, and macro viruses embedded in documents. While sandbox-based analysis has proven effective in enterprise environments, existing systems are resource-heavy and unsuitable for mobile devices commonly used by children. As a result, young users remain vulnerable to malicious documents shared through messaging platforms.

The absence of lightweight, child-friendly security solutions creates a serious gap. Parents and teachers cannot constantly monitor children's online activities, and children often lack the technical knowledge to recognize harmful files. This lack of protection increases the risk of exposure to malware, which may compromise personal devices, steal sensitive data, or enable further exploitation.

To address this problem, there is a growing demand for a parental-control–oriented mobile application that can analyze shared documents before children open them. By allowing users to upload or forward files into a sandbox environment, the system can simulate document execution, extract behavioral features, and classify files as safe or malicious. Alerts will then guide users: safe files can be opened, while malicious files trigger warnings to prevent harm.

The proposed research therefore seeks to bridge this gap by integrating sandbox-based behavioral analysis with machine learning into a lightweight mobile framework. Unlike existing approaches, it directly targets child safety in messaging platforms, empowering parents with a reliable control mechanism and ensuring that children are shielded from malicious documents during online communication.

II. OBJECTIVES

III.1 Main Objective

The primary objective of this research is to design and develop a sandbox-based behavioral detection system for identifying malware-infected documents shared through messaging platforms such as WhatsApp and Telegram. The proposed system will provide a lightweight and user-focused mechanism that executes files in a controlled environment before they are opened. In doing so, it aims to strengthen child protection by supporting a parental-control–oriented mobile application that ensures safer use of messaging platforms.

III.2 Sub-Objectives

- Review existing approaches: Study current malware detection methods, with emphasis on both static and dynamic analysis, and identify their limitations when applied to messaging platforms and mobile environments.

- Sandbox design and implementation: Develop a sandbox environment that can safely execute document files (PDFs, DOCs, and APKs) in isolation while capturing behavioral features such as file modifications, registry activity, API calls, and network communication.

- Machine learning classification: Build and train machine learning models capable of classifying documents as malicious or benign, using features extracted from sandbox execution logs together with external threat intelligence datasets.

- Mobile application integration: Incorporate the detection system into a mobile application that allows users to upload or forward files for verification, providing alerts that clearly indicate whether a document is safe to open or should be blocked.

- Lightweight optimization: Ensure that the system operates efficiently on mobile devices by

optimizing for low resource consumption, so that it does not disrupt usability or device performance.

- Evaluation and validation: Assess the system in terms of accuracy, precision, recall, and resource efficiency. Confirm that it meets the practical requirements of child protection and parental control within real-time messaging environments.

III. METHODOLOGY

Figure 1 - System Diagram

The proposed system is designed to protect children from opening harmful documents received through messaging platforms such as WhatsApp and Telegram. Parents can install and activate the app before handing the mobile phone to their children, ensuring that all documents are checked automatically before being opened.

1. Document Reception – When a child receives a document in WhatsApp or Telegram, instead of opening it directly, the system provides a floating UI button labeled *"Check Safety Before Opening"*.
2. Forwarding to Security App – If the child taps this button, the document is sent to the security application for inspection.
3. Quick Static Analysis – The system performs a fast scan to check for suspicious patterns, such as embedded macros, unusual file headers, or hidden executable content. This helps to filter obvious threats quickly.
4. Sandbox Execution and Monitoring – If the file still appears suspicious, it is opened inside a sandbox environment. In this controlled setup, the app monitors API calls, registry modifications, file system changes, and network activity to detect hidden malicious behaviors.
5. Machine Learning Classification – Behavioral features captured from the sandbox are analyzed using a trained machine learning model. The model decides whether the file is safe or malicious.
6. Alert System – Based on the classification:
   - Safe File → The system shows a green "Safe to Open" alert, and the child can open the file normally.
   - Suspicious File → The system shows a warning alert. The child then has two options:
     - Do not open the file.
     - Open the file in a Secure Viewer (inside the sandbox) so it cannot harm the device.
7. Child-Safe Usage – Since the parent activates the app beforehand, the child does not need technical knowledge. The system works automatically in the background and ensures that

dangerous documents are blocked or isolated.

This methodology combines floating UI access, static analysis, sandbox-based behavior monitoring, and machine learning classification to provide lightweight, real-time protection on mobile devices without needing a separate parental interface.

III.1 Requirement Gathering

The requirement gathering process combined a study of existing literature, analysis of current malware detection practices, and consultation with professionals. This ensured that the proposed solution is grounded in both academic research and practical needs.

- **Extensive Literature Review**: A systematic review of recent studies and journal articles was carried out on malware detection in documents, sandbox-based behavioral analysis, machine learning classification, and parental control applications. This review helped identify existing detection techniques, their limitations, and areas for improvement, particularly within messaging environments such as WhatsApp and Telegram.
- **Stakeholder Consultation**: Discussions and interviews were conducted with cybersecurity experts, mobile application developers, and child protection specialists. Their input provided practical insights into usability requirements, real-world malware threats, and the technical challenges of running sandbox mechanisms on mobile devices without reducing performance.
- **System Requirements Analysis**: Both functional and non-functional requirements were documented. Functional needs include sandbox integration for document execution, accurate malware classification, and real-time alerting. Non-functional needs cover performance optimization, resource efficiency, and child-safety features suitable for a parental-control–oriented mobile app. This analysis guided the design of a system that is secure, efficient, and appropriate for mobile use.
- **Use Case Scenarios**: Practical scenarios were created to validate the system design. For instance, when a child receives a suspicious PDF in WhatsApp, the file is forwarded to the security app. The app executes the file in a sandbox, analyzes its behavior, and classifies it as safe or malicious. An alert is then displayed to both the child and the parent, ensuring that harmful files are blocked while safe documents remain accessible. These scenarios confirm the system's relevance and effectiveness in protecting children and general users from document-based malware.

III.3.1 Past Research Analysis

The review of earlier studies provides important insight into how document-based malware has been addressed in different contexts of messaging and file sharing. At the same time, it shows where existing approaches fall short, especially when applied to mobile platforms such as WhatsApp and Telegram.

III.2.2-i Literature Review of Key Studies:

- Research [1]: Concentrated on static signature-based detection of malicious PDF and DOC files. While the method worked for known malware, it performed poorly against polymorphic or obfuscated threats.
- Research [2]: Used sandboxing techniques to detect malicious scripts in office documents. The study, however, was limited to desktop environments and did not consider use on mobile devices or in messaging platforms.
- Research [3]: Proposed machine learning approaches for detecting macro viruses within documents. These models, though effective in controlled datasets, lacked validation in real-world messaging environments such as WhatsApp or Telegram.
- Research [4]: Examined parental control applications designed for children's online safety. Most of these tools concentrated on web filtering and activity monitoring, with little focus on sandbox-based malware analysis for shared documents.

III.2.2-ii Identification of Gaps

- Integration of sandbox-based behavioral analysis in messaging platforms: Although sandboxing is common in enterprise security, few studies have applied it directly to mobile messaging apps where document sharing is widespread.

- Multi-file malware detection: Previous research often concentrates on a single file format, such as PDFs or Word documents. There is little evidence of frameworks able to analyze multiple file types (PDFs, DOCs, and APKs) in one system.

- Lightweight mobile implementation: Existing sandbox systems are usually resource-intensive, making them impractical for smartphones. Very few studies address lightweight designs that can balance detection accuracy with mobile device performance.

- User-centric and child-safety context: Prior work has not given adequate attention to workflows that let users or parents decide which documents to analyze before opening. Nor has it targeted protection specifically for children using messaging platforms.

III.2 Feasibility Study

The feasibility of this research is assessed by examining the tools, technologies, and analytical methods needed for the development and implementation of a sandbox-based malware detection system in messaging platforms. The study considers both the availability of resources and the suitability of the chosen approaches for deployment on mobile devices.

III.3.2 Technical Feasibility

III.2.2-i Tools and Technology

- **Python**: Acts as the primary programming language for implementing machine learning models, preprocessing pipelines, and integration with sandbox logs. Widely supported libraries make it ideal for behavior-based malware detection research.
- **Flutter Framework**: Used to develop a cross-platform mobile application (Android & iOS) that provides the user interface for uploading/forwarding suspicious documents. Flutter ensures lightweight performance and consistent design across both operating systems.
- **Cuckoo Sandbox / Any.Run**: Provides isolated execution environments for suspicious documents (PDFs, DOCs, APKs). These tools capture behavioral data such as API calls, file modifications, registry activity, and network traffic during execution.
- **Scikit-learn & TensorFlow/PyTorch**: Libraries to build and train machine learning classifiers for identifying malicious vs. benign files. TensorFlow/PyTorch will handle deep learning experiments if required, while Scikit-learn supports traditional algorithms such as Random Forest or SVM.
- **SQLite / Firebase**: Databases to log results, store sandbox analysis metadata, and

support real-time communication between the mobile app and the backend. Firebase also enables cloud-based updates for threat intelligence.

- **Docker**: Used to containerize the sandbox environment, ensuring safe execution of suspicious files without affecting the host system. This also allows reproducibility and scalability during testing.

III.2.2-ii Algorithm and Architectures

- **Sandbox-Based Dynamic Analysis**: Suspicious files are executed in a controlled sandbox, capturing runtime events such as system calls, process creation, registry changes, and network activity.
- **Machine Learning Models:**
    - Traditional ML models: Random Forest, Support Vector Machine, Logistic Regression.
    - Lightweight Neural Networks: Optimized for mobile-based deployment.
      These models classify documents as benign or malicious using sandbox-derived features.
- **Hybrid Detection Architecture:** Combines static pre-checks (macro detection, suspicious metadata, obfuscated scripts) with dynamic analysis to improve detection accuracy and reduce unnecessary sandbox executions.
- **Mobile Application Architecture:** A parental-control–oriented security app built in Flutter. The app allows users to forward suspicious files, view safety alerts, and optionally open risky documents inside a secure in-app sandbox viewer.

III.2.2-iii Techniques

- **Data Preprocessing**:
  - Cleaning sandbox logs and normalizing extracted features.
  - Feature engineering from behavioral indicators (e.g., frequency of API calls, abnormal network requests).

- **Sandbox Simulation**: Execution of files in isolated environments to log runtime behaviors. Multiple sandbox runs may be performed to detect evasion attempts.

- **Behavioral Feature Extraction**: Identifying patterns such as unusual registry modifications, persistent processes, or command-and-control traffic.

- **Classification & Evaluation**: ML models trained and validated using datasets of both benign and malicious files. Metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC will be used.

- **Performance Optimization**:
  - Lightweight model deployment (using TensorFlow Lite for mobile).
  - Efficient resource handling so the app runs smoothly without draining device battery or storage.

- **User Alert & Logging**: Clear visual alerts (e.g., *Safe to Open* or *Malicious File Detected*) with secure sandbox-viewing options. Logs are securely stored for parental review and future retraining of ML models.

III.3 PROJECT REQUIREMENTS

III.3.1 Functional Requirements

- Document Input Handling: The system should allow users to upload or forward documents (PDFs, DOCs, APKs) directly from messaging apps such as WhatsApp or Telegram into the security application for analysis.

- Sandbox Execution: Files must be executed in a secure, isolated environment where their runtime behavior can be observed without risking the user's device.

- Behavioral Feature Extraction: The system should record relevant behavior during execution, such as API calls, registry changes, file system actions, and network requests, for later classification.

- Malware Classification: A machine learning model must analyze the extracted behavior and classify the document as either benign or malicious.

- Alert System: After classification, the system should generate a clear result. If the file is safe, the user is notified and allowed to open it. If the file is malicious, the system should issue a warning and block the file or allow secure viewing.

- Parental Interaction (Optional): The system may support parental controls that notify a parent when a potentially dangerous file is detected on the child's device.

III.3.2 Non-Functional Requirements

- Accuracy: The malware detection system should offer a high level of precision and recall, ensuring that both false positives and false negatives are minimized.
- Lightweight Performance: The application must be optimized to run on mobile devices without noticeable delay or excessive use of battery, memory, or storage.
- Usability: The user interface must be simple and accessible, especially for children and non-technical users. Prompts and alerts should be clear and actionable.
- Security: The sandbox environment and app must protect user data and ensure that potentially malicious files cannot escape the analysis container.
- Compatibility: The system should support common Android versions and be capable of processing widely used document formats without compatibility issues.
- Maintainability: The architecture should support updates to machine learning models and threat data so the detection system can adapt to evolving threats.

III.4 WORK BREAKDOWN AND GRANT CHART

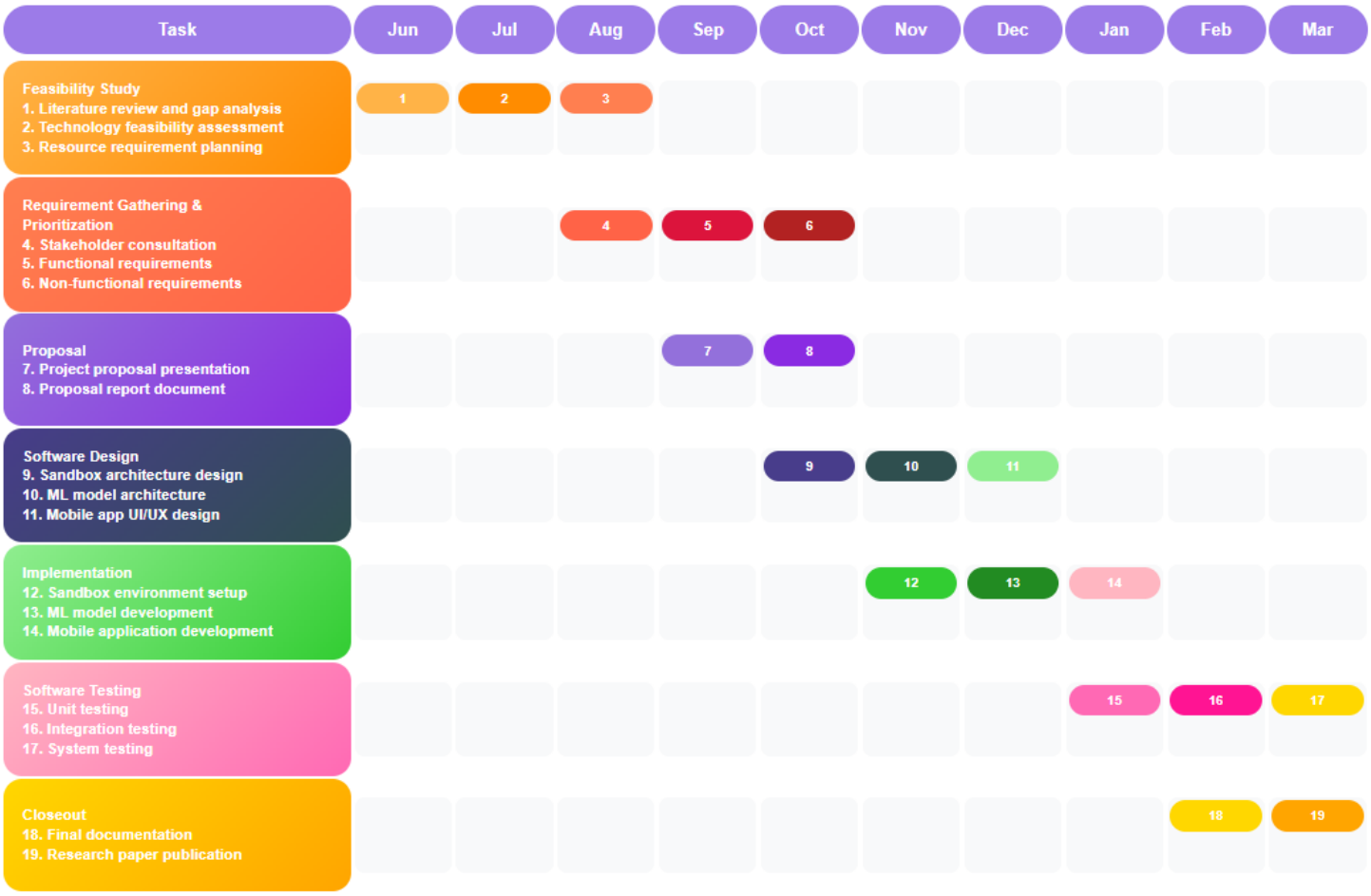

Figure 2 – Grant chart

Figure 3 - Work Breakdown Chart

III.5 COMMERCIALIZATION

The proposed sandbox-based malware detection system has strong potential for commercialization as a mobile security solution tailored to modern messaging environments. With the growing use of WhatsApp, Telegram, and similar platforms by young users, there is an increasing need for simple, reliable tools that protect children from harmful content—particularly document-based malware. This solution addresses that need through a lightweight mobile application that scans shared documents in real time and alerts the user or parent before harm occurs.

The core innovation lies in combining sandbox-based behavioral analysis with machine learning, packaged in a format suitable for mobile users. Unlike general antivirus apps, this system is built around the specific threats seen in messaging apps and allows users to make decisions before opening a file.

The mobile app could be positioned as a security layer for parents, schools, and education-focused mobile device programs. It may also appeal to general users seeking document-level protection from phishing or malware attacks spread through chat applications.

Key Commercialization Points:

- Target Market: Parents of school-age children, school IT administrators, family device management tools, mobile device manufacturers, and secure messaging platform vendors.
- Unique Selling Proposition: A child-focused document malware detection app that uses real-time behavioral analysis rather than basic static scanning. It allows safe document viewing without blocking legitimate files needed for learning or communication.
- Revenue Model: The app could follow a freemium model, offering basic protection features for free and charging for premium options such as parental reports, advanced analytics, cloud sandboxing, or multi-device control. It could also be licensed to

educational institutions or bundled with child-focused device setups.

- Market Entry Strategy: Early adoption can be driven through partnerships with schools, child-safety advocacy groups, and mobile OS vendors. Showcasing use cases where children are protected from document-based threats in real-world scenarios will help build trust and visibility.

In summary, as digital learning and communication expand among young users, a lightweight mobile solution that protects against document-based threats in messaging apps meets a growing need. Its focused design, behavioral detection accuracy, and usability give it strong commercial potential in the family cybersecurity and education technology sectors.

IV. BUDGET AND BUDGET JUSTIFICATION

As the proposed research is software-oriented, no physical hardware is required. The core development involves sandbox analysis, machine learning, and mobile app integration, all of which can be carried out using open-source tools and university resources. The main expenses relate to computing infrastructure, internet access, publication, and essential research materials. A summary of the estimated costs is provided below.

Table IV-1 - Budget Analysis

Type	Description	Cost
Internet and web browsing	For downloading tools, research, communication, and system testing	5000 LKR
Cloud sandbox / server use	For simulating malware behavior and testing dynamic analysis	8000 LKR
Publication Cost	For submitting research to a journal or conference	15000 LKR
Stationary	For notes, sketches, rough reports, and final documentation	4000 LKR
Android test device (optional)	For testing app performance on a physical device (if not already available)	5000 LKR
Google Play developer account	One-time fee to publish the mobile app on the Play Store	8000 LKR
Total		45000 LKR

The internet and browsing costs support continuous research, software updates, and communication. Cloud sandbox and server costs cover limited runtime simulations for testing malware behavior. Stationery and printing expenses support note-taking, report drafts, and reference material during the study. Publication fees account for submitting the final paper to a conference or journal.

V. Summary

This project proposes a lightweight, mobile-based malware detection system designed to protect users—particularly children—from document-based threats shared through messaging platforms such as WhatsApp and Telegram. The system uses sandbox-based behavioral analysis to execute and observe files in a secure environment, capturing runtime behaviors such as file changes, registry activity, and network communication. These features are then analyzed using machine learning models to classify files as safe or malicious. The application allows users to forward suspicious files for analysis and receive alerts before opening them. In cases where a file is needed despite risk, the system offers a secure viewing option. The project includes a review of current detection methods, identification of research gaps, and validation through prototype testing. A commercialization plan targeting parents, schools, and child-safety organizations is also outlined to support broader adoption.

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